

Karan Gupta

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EDUCATION

University of Waterloo

Bachelor of Applied Science in Computer Engineering, Minor in Economics

Expected June 2030

GPA: 3.95

EXPERIENCE

Software Development Engineer Intern

May 2026 – Present

Amazon

Toronto, ON

- Reduced legacy service migration time by 98% (2 weeks → 1 hour per stage) by building multi-step AI agents using prompt chaining and CoT reasoning to automate Java architecture transformation across thousands of lines.
- Architected an agentic RAG pipeline with few-shot prompting to achieve 92% translation accuracy across legacy API calls, algorithm structures, and AWS cloud deployment patterns into modernized service frameworks
- Developed a pluggable natural language interface enabling on-demand algorithm adjustments during automated migration, allowing seamless integration with modern service handlers without manual intervention

Software Engineering Intern

January 2026 – April 2026

Manulife Financial Corporation

Waterloo, ON

- Reduced system downtime risk by scripting a New Relic data exporter and architecting NRQL queries, enabling early detection of 3+ resource bottlenecks before system failure
- Designed a Python analysis script and SQL telemetry pipeline to process clickstream data, identifying 20+ suspicious user anomalies by applying mathematical modeling and cross-referencing behavioral logs.
- Implemented 15+ Salesforce features to automate data entry during document uploads for specialized lending workflows, reducing manual input for high-value client processing using Apex and LWC

Autonomy Software Developer

Sep. 2025 – Present

Waterloo Aerial Robotics Group

Waterloo, ON

- Improved real-time signal detection accuracy by 13% using OpenCV2 computer vision algorithms through iterative parameter tuning and validation
- Built multi-process telemetry and command systems in Python (PyMAVLink) to simulate UAV communication, reducing message latency by 30% across distributed processes

PROJECTS

Reparo (Hack Canada 2026 Winner) | *Python, Gemini API, React, Node.js, Swift, PyTorch*

- Built an agentic AI system using Gemini vision models for product classification and reparability assessment with 90%+ accuracy.
- Reduced user search time for replacement parts by 70% and enabled access to 1,000+ real-time listings by integrating Shopify Storefront API and SerpAPI to automate matching and streamline repair-to-checkout flow
- Won 1st place in Reactiv Track (\$5,000) at Hack Canada 2026; shortlisted for Most Complex AI Hack

DeliriumWatch | *Raspberry Pi, Arduino, Python, OpenCV, HTML, CSS, Flask, C/C++*

- Built a real time Python monitoring pipeline with SQL-backed secure login and role-based access control for hospital staff, resulting in over 90% reduction in manual environmental monitoring during simulated testing.
- Implemented OpenCV eye detection and live Flask dashboard to surface abnormal patient conditions in real time, reducing average nurse response time by 40% during simulated testing

TailorAI (Silhouette) | *Python, TensorFlow, scikit-learn, OpenCV, Node.js, React, MongoDB*

- Built a full-stack AI platform using computer vision and the MERN stack, processing 200+ images to automatically extract 10+ body measurements per user and eliminating reliance on manual sizing inputs
- Implemented a Python CV pipeline with landmark normalization using scikit-learn and TensorFlow, improving statistical measurement consistency by 40%
- Designed scalable MongoDB schema and REST APIs supporting 150+ items with sub-200ms query latency.

SKILLS

Languages: Python, Java, C, C++, Swift, TypeScript, JavaScript, SQL, HTML, CSS

Frameworks & Libraries: PyTorch, TensorFlow, scikit-learn, React, Node.js, Flask, FastAPI, OpenCV2, PyMAVLink, Dagger

Developer Tools: Git, Docker, AWS (S3, SQS, SNS), Postman, Kubernetes, Xcode